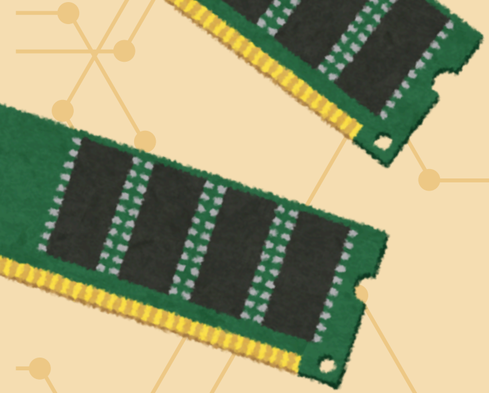
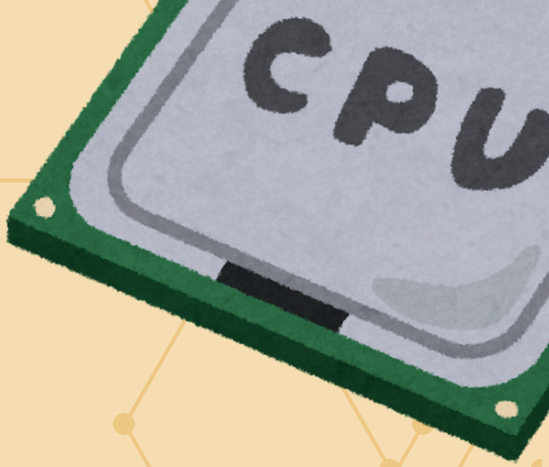


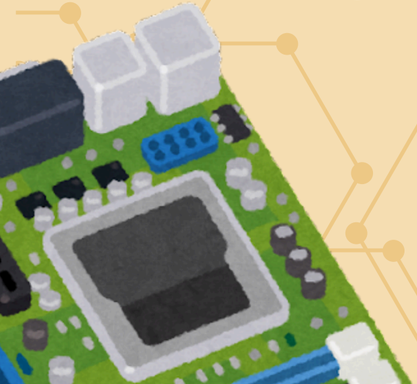


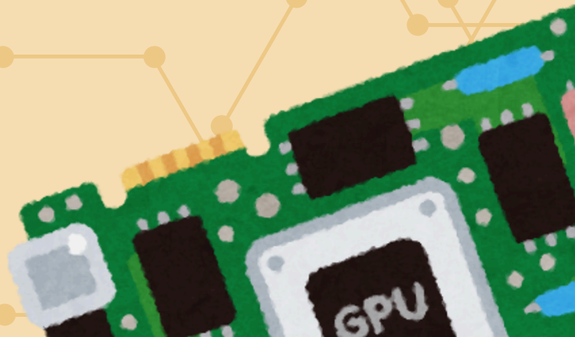
CPE 312 - CPE31S3

Voice-Based Gender Classification using Machine Learning

Group 1

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Garcia, Ron Allen
Tiong, Martin Andrei





Introduction

HUMAN SPEECH...

- Communication
- Expression
- UNIQUE IDENTIFIER

THE PROJECT...

Specialized machine learning model for gender classification

IT'S CHARACTERISTICS...

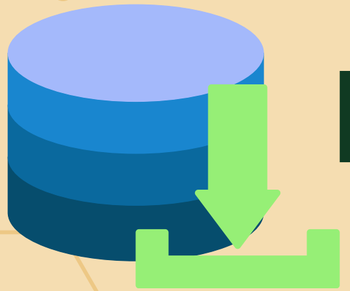
Evaluates voices by extracting pitch, frequency, and other acoustic properties

THE GOAL

Gender classification model for human to computer interaction medium

Methodology

Data Acquisition



- “Hello” voices from movie clips from YouTube
- librosa to get average pitch (pyin function)

Data Cleaning



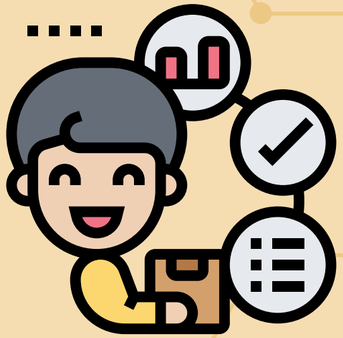
- Cleaned missing values
- Added gender column (for gender distribution)

Model Training



- Logistic regression method
- 70/30 split

Model Evaluation



- Model accuracy
- Classification report
- Confusion matrix
- Test model with a new voice data

Results

```
# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy:.2f}')
```

```
Accuracy: 0.83
```

```
# Display classification report
print('\nClassification Report:')
print(classification_report(y_test, y_pred))
```

```
Classification Report:
              precision    recall  f1-score   support

      female       0.85      0.82      0.84        28
        male       0.81      0.85      0.83        26

    accuracy                           0.83        54
   macro avg       0.83      0.83      0.83        54
weighted avg       0.83      0.83      0.83        54
```

Model Accuracy

- Strong feature identification between male and female voices
- Good results for a univariate analysis

Precision

- 85% - predicting female voices
- 81% - predicting a male voices

Recall

- 82% all actual female voices
- 85% all actual male voices

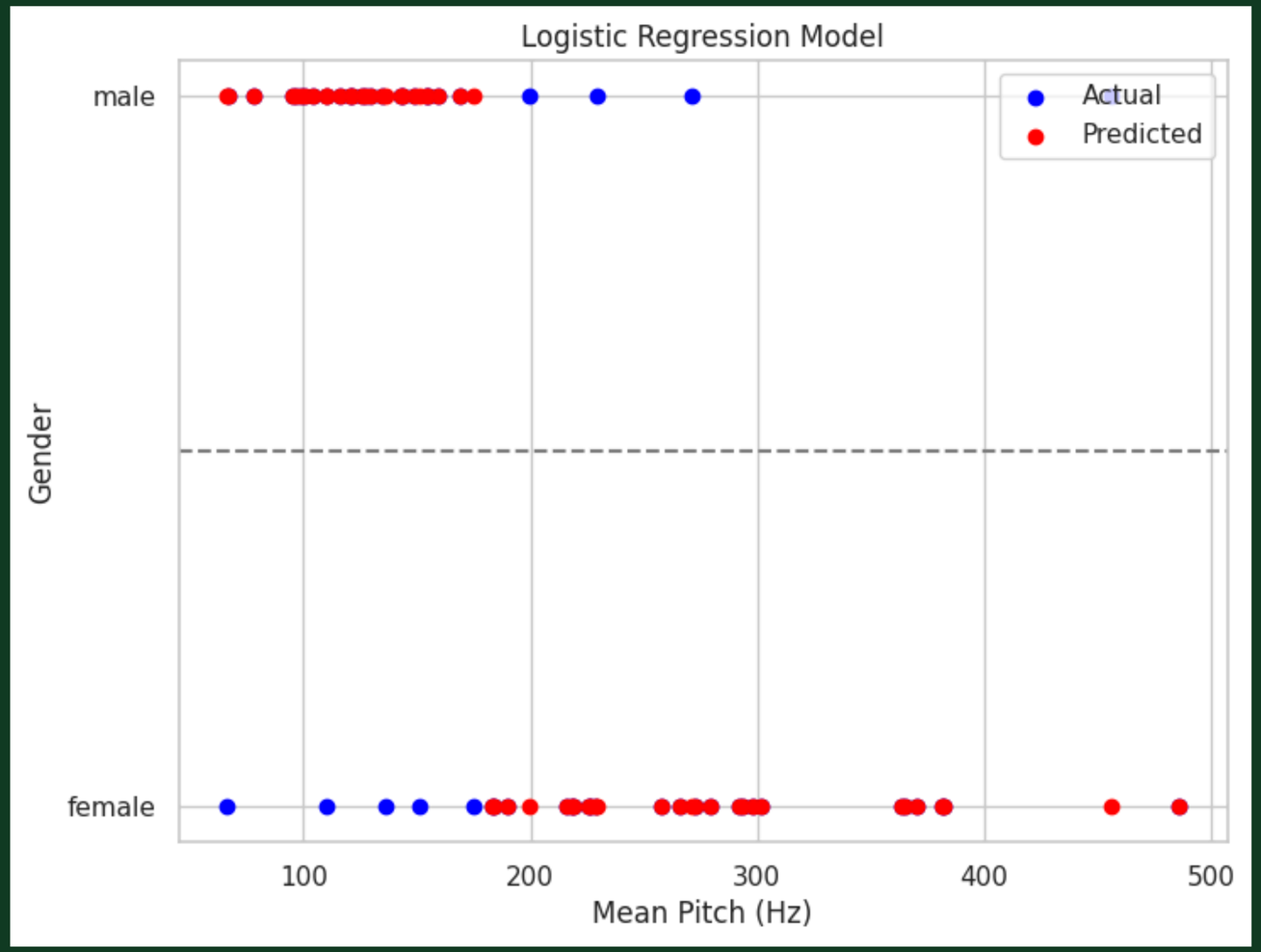
F1-Score

- Balanced performance

Support

- 26 actual 'male' voices
- 28 actual 'female' voices
- Balanced test data distribution

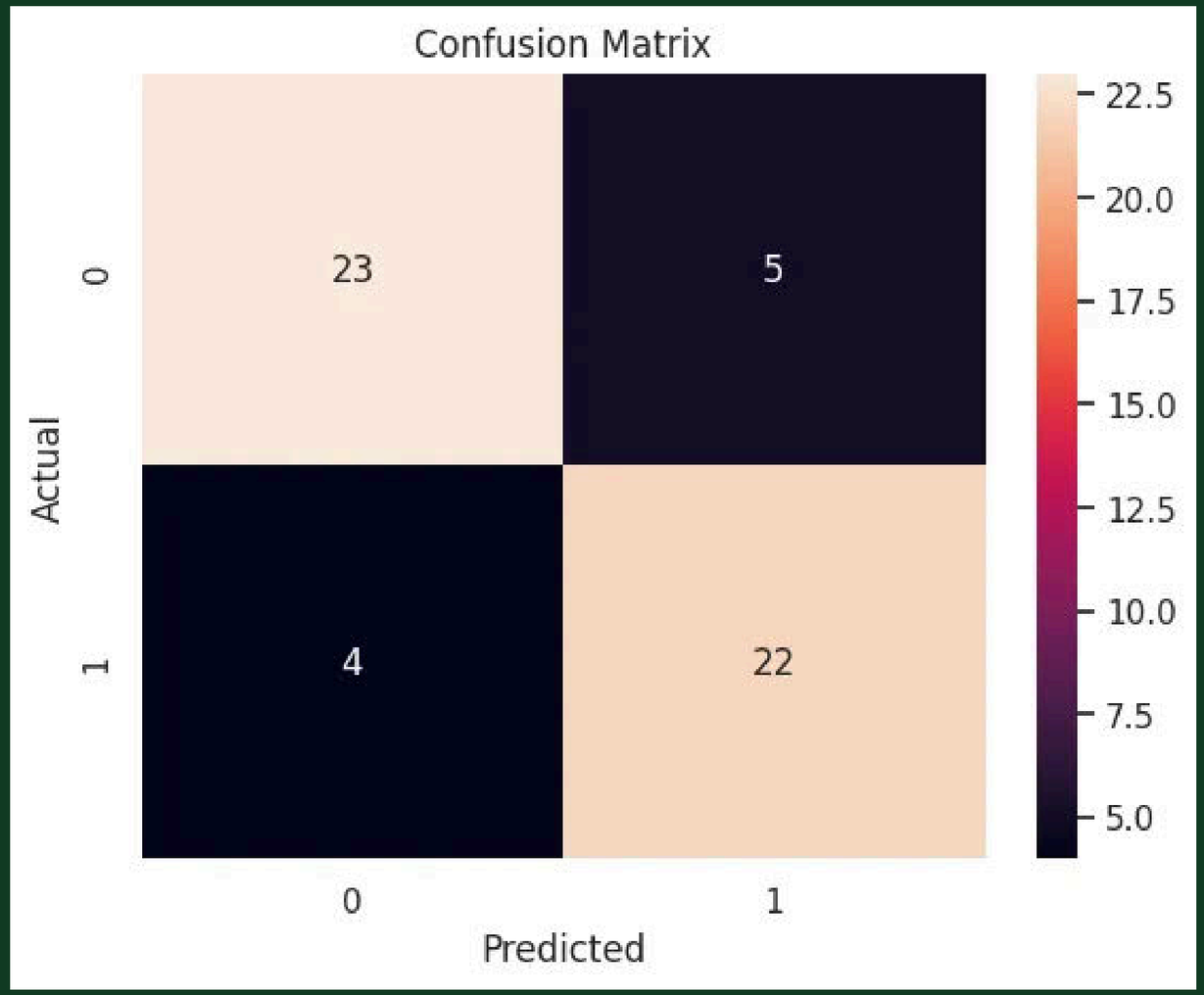
Results



Logistic Regression Graph

- Male voices = lower mean pitches, female voices = higher pitches
- Misclassifications & model performance

Results



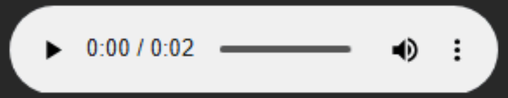
Confusion Matrix

- Performs well for both genders
- female voices had slightly higher precision, male voices had slightly higher recall

Results

```
# Provide path for new audio file
new_audio_path = '/content/drive/MyDrive/new_voices/aldous_voice.mp3'
```

```
if new_audio_path:
    display(ipd.Audio(new_audio_path))
else:
    print("No audio file path provided.")
```



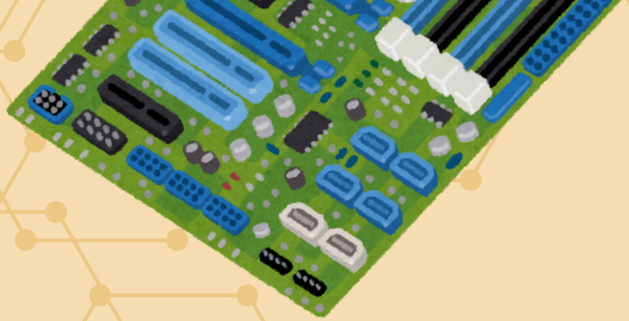

```
...  Predicted Gender: MALE
     Mean Pitch (Hz): 78.21
```

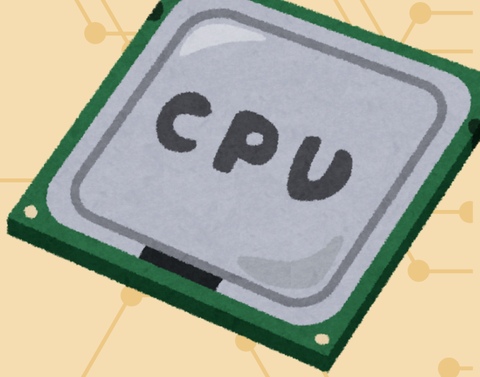
Recommendations

- More target variables for better reference
- Better ways to manipulate extracted data
- More diverse set of voices

References

- Ertam, F. (2019). An effective gender recognition approach using voice data via deeper LSTM networks. Applied Acoustics, 156, 351–358. https://doi.org/10.1016/j.apacoust.2019.07.033
- Livieris, I. E., Pintelas, E., & Pintelas, P. (2019). Gender Recognition by Voice Using an Improved Self-Labeled Algorithm. Machine Learning and Knowledge Extraction, 1(1), 492–503. https://doi.org/10.3390/make1010030
- Nandita and B. Jain, "A Comprehensive Review of Machine Learning Techniques for Voice-Based Gender Recognition," 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), Delhi, India, 2023, pp. 1-7, doi: 10.1109/ICCCNT56998.2023.10306939.
- Spiceworks. Everything You Need to Know About Logistic Regression. (2022, April 8). Spiceworks Inc; Spiceworks. https://www.spiceworks.com/soft-tech/what-is-logistic-regression/





Thank you!

